

Stat 248: Final Project

Spring 2026 – Liberty Hamilton

Overview

The final project is your opportunity to apply the methods from this course to a real scientific or analytical question using time series data. You will identify a question in your research area (or a domain that interests you), obtain appropriate data, conduct a time series analysis, and communicate your findings in both a written code report and an oral presentation.

Final projects will be completed and submitted individually, not as a group project.

Your choice of methods should be driven by the structure of your data and the demands of your question. Don't just provide a methods demo (as in previous labs, for example). The goal is to try to answer a true scientific or analytical question. I am excited to see what you come up with!

Assignment Components and Deadlines:

1. Project Proposal: Due April 16, 2026 by 11:59pm

This will consist of a 1–2 page document containing the following (please use headers!):

The question. State a specific scientific or analytical question. A good question might be "Does continuous heart rate data exhibit periodicity beyond a 24-hour circadian cycle, and does that periodicity differ between clinical and control groups?". A bad question (or at least, underspecified), would be "I will analyze heart rate data." Your question should be precise enough that someone in your field would recognize it as meaningful, and someone outside your field could still understand what a "yes" or "no" answer would look like.

Description of the data. Describe the dataset you will use: what it measures, where it comes from, the temporal resolution, the number of observations, and the number of distinct time series (if applicable). You should have the data in hand by the proposal deadline. Proposals with vague data plans will be returned for revision.

Why this is a time series problem. In one paragraph, explain what temporal dependence structure you expect in the data and why ignoring it would lead to incorrect or incomplete answers to your question.

Proposed methods. Identify 2–3 methods from the course that you plan to use and explain why they are appropriate for your question and data. You should use at least two substantive analytical methods — plotting the ACF and fitting a single ARIMA model does not constitute a complete analysis.

Expected outcomes. Describe what a positive result would look like and what a null result would look like. It does not matter if your expected outcome turns out to be true! I just want you to think about what you might see and how you would interpret it.

I will review proposals and provide feedback on scope, feasibility, and methodological fit. I will ask you to revise if the question is too vague, the data is insufficient, or the proposed methods don't match the question. Proposals will be approved by April 21.

2. Final Presentation Video: due May 14, 2026 by 11:59pm

A recorded 20 minute oral presentation, structured as:

- **Introduction.** What is the problem and why does it matter? Assume your audience is technically competent but not expert in your specific domain. Give enough background that they can follow your reasoning.
- **Data and methods.** What data did you use and what did you do with it? This should include enough detail that someone could evaluate your choices, but it should not be a deep dive into your code. Equations and diagrams are helpful.
- **Results.** What did you find? Present your results clearly, with appropriate figures. Interpret them in the context of your original question.
- **Conclusion.** What do your results mean? What are the limitations? What would you do next with more time?

3. Code Report: due May 14, 2026 by 11:59pm

A reproducible analysis (Jupyter notebook, R Markdown) that includes:

- A brief written narrative connecting the code to the analysis (not just raw code cells).
- All preprocessing, analysis, and visualization code needed to reproduce your results from the raw data.
- Model diagnostics: residual analysis, stationarity checks, goodness-of-fit assessments, or other validation appropriate to the methods used.
- Clear comments and organization. Someone familiar with the methods should be able to follow your analysis without your oral explanation.

If possible, include the data with your submission, or provide instructions for accessing it if it is too large or access-restricted. If it is restricted (e.g. subject to HIPAA restrictions, needs IRB approval), please provide a clear reason and try to include simulated data to test the code.

Grading

You will be graded on the following:

- **Problem Formulation and Motivation:** Do you have a clearly articulated question? Have you expressed why your question matters and why time series methods are necessary to answer it?
- **Methodological Appropriateness and Execution:** Did you choose methods that are well-suited to your data and question? Did you check and discuss the assumptions for those methods and perform appropriate diagnostics?

- **Results and Interpretation:** Can you explain what you found in terms that connect back to your original question, not just in model coefficients?
- **Code Quality and Reproducibility:** Is your analysis reproducible, organized, and documented well enough that a reader can follow it?
- **Presentation:** Did you communicate your problem, approach, and findings clearly within the time limit? Could someone outside your field understand your talk?